Hybrid Machine Learning and Queuing Theory Approach for Real-Time Hospital Resource Allocation

K. Sasikala
Department of Mathematics,
Dr SNS Rajalakshmi College of Arts and Science (Autonomous),
Coimbatore, Tamil Nadu, India
sasikala1301@gmail.com

S. Karpagalakshmi
Department of Commerce
Dr. SNS Rajalakshmi College of Arts and Science (Autonomous)
Coimbatore, Tamil Nadu, India
karpagambaburaj@gmail.com

M. Rajeswari ,
Department of Computer Science with Data Analytics.
Dr SNS Rajalakshmi College of Arts and Science,
Coimbatore, Tamil Nadu, India
lidishraji@gmail. Com

C Abinaya
M.Sc. Mathematics
Dr SNS Rajalakshmi College of Arts and Science ,
Coimbatore, Tamil Nadu, India
abinayachandran21@gmail.com

E Janani ,
M.Sc. Mathematics ,
Dr SNS Rajalakshmi College of Arts and Science ,
Coimbatore, Tamil Nadu, India
easwaranjanani21@gmail.com

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***Abstract*-- Healthcare resources are dispersed in accordance with established norms. However, patients enter the system in an unpredictable manner. It causes healthcare inefficiencies, which must be addressed. As a result, solutions for simply satisfying demand, such as resource use patterns, must be developed. Traditional hospital queuing and forecasting systems have been ineffectual. The results are disappointing in two ways: there are either too few or too many patients. Both waste hospital resources. To address these concerns, a hybrid machine learning-queueing theory framework was developed. The proposed system forecasts workloads and employs hybrid queue modeling and real-time optimization to change elastic queue sizes as needed. It uses a K-SVR prediction model and an RL dispatcher to make real-time bed and personnel allocation decisions. Forecast outputs are updated periodically throughout the process to adjust and balance queue parameters. The study results, which preserve privacy, show a 31.8% decrease in patient wait time and a 21.9% decrease in unoccupied beds. The forecast error (MAPE) is about 1.2% lower than the same organization's queuing and single ML model. The proposed method is both cost-effective and scalable, relying on neural networks and machine learning for real-time, adaptive hospital resource distribution.**

Keywords: Hybrid Model, Queuing Theory, Hospital Resource Management, Real-Time Allocation, Bed Utilization, Healthcare Optimization, Dynamic Scheduling, Resource Forecasting.

I. INTRODUCTION

Hospital resource allocation has always been a source of worry in health care, due to constantly changing patient demand and a myriad of constraints such as personnel, material, space, and capacity. Hospitals are regarded as complex service systems in which numerous interdependent resources (such as beds, personnel, equipment, and departments) vary stochastically over time [1]. Real-time fluctuations in patient arrival and service time are challenging to manage with typical decision-making systems based on scheduled numbers, and queues offer an oversimplified solution. On the one hand, machine learning algorithms have improved accuracy in predicting patient queues and bed occupancy; nevertheless, these forecasts are rarely related to real-world operational levers or clinical value [2]. A separation of prediction and planning leads to inefficiencies like as wait times, surges, and action lags, all of which contribute to uneven workloads and the requirement for extra capacity. Addressing such inefficiencies will demand the creation of a cognitive system with predictive and responsive skills, as well as real-time decision-making abilities [3]. A real-time responsive empirical resource management system is only achievable with a combination of machine learning and queuing theory, which is a significant invention capable of continuously calibrating demand and supply in hospital resources [4]. The difficulty of providing timely, high-quality care with limited resources remains a major concern for healthcare systems around the world [5]. Demand surpassing capacity forecasts hospital problems during emergencies, seasonality, and unexpected pandemics.

A manual or rule-based allocation method is simply too stiff to handle such dynamic situations. Traditional models, on the other hand, serve as the foundation for flow regulation, but they assume fixed rates for arrivals and services. However, most of these ML techniques are not designed to handle dynamic assignment. It will be corrected as machine learning techniques are integrated with queuing theory. The hypothetical system must be able to refer back to prior data. It would need to use the results of its predictive model to change request quantities 'on the fly', making hospital operations more proactive rather than reactive. The main selling feature of this hybrid framework has been its capacity to build a functioning hospital resource allocation system in which decisions are both reasonable and timely. It contains the premise that intelligent prediction modeling, combined with operationalized optimization, creates a connected, continuous process that learns and adjusts.

All combined estimate hourly patient demand using regression and auto regressive moving average (ARMA) models; with predictions for the parameters of a queuing model based on this forecast demand, classic timeliness models will direct control; and an RL model will run continuously in real time to generate dispatch recommendations. It can also use admission and discharge data from existing EMRs (essentially sparse sensor readings) to create a situationally aware real-time decision support system for HCSC hospitals and clinics. Second, it must be scalable over time, allowing hospital departments to meet current and future demands without hiring additional personnel. The proposed method seeks to maximize bed usage, reduce patient wait times, enhance staff conditions, and plan for peak demand or disaster in the event of an operational breakdown. The study intends to enhance the field of healthcare analytics by presenting a novel hybrid-knowledge system in which data-driven machine learning predictions are linked to control based on queuing theory derived from operational evidence. Part of the process is assessing demand, and part includes ensuring that demand is met during allocation decisions. As a result, the framework is flexible enough to interact with existing hospital sustainability management systems. Finally, because of its ability to adapt dynamically to random fluctuations in patient flow, the proposed system outperforms existing optimizations achieved under such conditions, bringing a compromise between traditional queuing system models and completely predictive systems. Section 2 presents the study's main findings, which include an analysis of past research on hospital resource management, machine learning for prediction tools, and queueing theory applications.

Section III describes the proposed system's design and operational processes. Section IV describes how data is processed, algorithms are executed, and optimization occurs. Section V presents data and result analysis, as well as comparisons to traditional model performance, application, and benefits. Section VI discusses first footholds and missing elements, as well as directions for future development, whether through challenge or by implying greater elasticity and fit into a broader range of venues when hospital care is required.

II. Related Work

Congested queuing networks are an important domain of application for integrated simulation and machine learning-based methods, and there is rising interest in using such methods to improve the prediction of delays in congested systems as well as system responsiveness across healthcare operations. Such breakthroughs include a simulation-based technique in which, using the available information of the arrival process from our machine-learning algorithms, we can estimate the real-time state delays experienced by patients while simultaneously predicting the queue length online [6]. The reinforcement learning agents create rules for efficient dispatching by interacting with patient simulations as well as the task, environment, resource, and so on over several episodes. These adaptive rules adjust to changeable congestion and staff limits while adhering to a real-time ratio of patient throughput to service quality on systems [7]. Efficient triage enabled by queuing principles allows for the use of artificial intelligence and game theory; hybrid technology has switched from a narrow search for simply operational efficiency to clinical prioritization. The models directly face the ethical and pragmatic errors in treating a paucity of clinical resources PPEs in epidemic cases by implementing automatic fairness criteria and competing marketplace balancing with [8].

The study investigates the potential synergy of machine learning and simulation to inform scheduling and flow optimization in hospitals, assessing and adjusting these various scheduling configurations using forecast models of demand for care in conjunction with discrete-event simulation, before trajectories are established into practice [9]. Recent research has looked into the use of metaheuristic optimization tools, such as particle swarm optimization (PSO) and Markov decision models, to maximize resources for multiple patient classes at once. Stochastic state transitions are employed to characterize patient progress along care pathways, and optimization techniques are used to set service priorities so that queue delays at each level are minimized, as is idle time at service mechanisms. For example, because of PSO's adaptability, near-optimal queue configurations for different services in healthcare systems can be attained even in the face of high operational uncertainty, resulting in fair service provision to both essential and non-critical patients [10]. These systems adjust the bandwidth, processing capacity, and accessibility of medical equipment based on the expected patient volume and communication intensity. Combining real-time sensing and prediction algorithms will allow hospitals to build self-regulating infrastructures that can prioritize which emergency operations to perform [11]. Hospitals may be able to estimate patient peaks by combining historical data from prior outbreaks with models created to predict probable future epidemics. Adaptive storage can move resources from one department to another in hours without compromising procedures, even when demand increases unexpectedly [12].

The application of machine learning technologies to emergency transportation services improves ambulance allocation, routing, and demand forecasting. Its frameworks incorporate spatial-temporal elements, and its predictive algorithms aid in the dispatch process. All of these provided quick arrival, long-lasting transition, and comprehensive coverage with little additional effort [13]. Deep reinforcement learning has significantly improved outpatient scheduling for patients. For example, one model discovered that intelligent agents can use patient flow data to shorten wait times and improve service delivery between departments and physicians. The approach uses feedback loops and integrates many other variables into forecasts that doctors are unaware of. The rationale is really adaptable; it may be altered in response to repeated reappearances and no-shows, allowing the process to be more flexible than typical guideline-based frameworks [14]. Finally, they result in optimization-based hybrid models that combine queuing theory, mixed-integer linear programming, and adaptive simulated annealing to address the patient scheduling problem. These use deterministic and probabilistic search methods to optimize competing objectives including cost, time, and service equity. These systems have excellent convergence capabilities for scaling over multiple departments while maintaining a relatively constant preprocessing cost [15].

III. Proposed System

The proposed hybrid machine learning and queuing theory-based approach enables near-real-time resource allocation to hospitals while overcoming existing systems' inefficiency, stiffness, and inflexibility. Traditional queuing-based models, which are used to manage hospital beds and staff, assume static parameters, such as fixed arrival rates, service times, and department flow, making them unsuitable for the dynamic and stochastic nature of healthcare operations. Similarly, while machine-learning-based prediction systems in the healthcare domain may be useful for accurately forecasting patient inflow or bed demand in the medium to long term, these systems exist in virtual isolation from the real-time resource allocation process; they are simply predictive algorithms. It leads to a failure to match forecast and execution; projections may be accurate, but hospitals will still experience capacity misalignment, long wait times, and over- or underutilized resources. It, in turn, reveals limitations in more generic machine learning approaches than directly applying queuing theory, which may lack the generality required for widespread application, resulting in the proposed system that combines machine learning predictive accuracy with queuing theory logic to create a feedback-driven integrated framework. The main idea is to create a hospital system that can not only predict the demand for queue services in advance with high accuracy but also use those predictions to re-adjust potentially all parameters, including policies, to achieve optimal results instantly, resulting in both prediction accuracy and system adaptivity.

The suggested system is divided into four phases: data gathering, forecasting, optimization, and adaptive assignment. Phase 1 collects real-time data from hospital information systems on inpatient admissions, discharge hours, bed occupancy records, and staff availability. Data pretreatment techniques such as noise removal, normalization, and feature extraction are used to ensure that only valid data is sent into the predictive engine. It involves two steps. The second step employs a combined ensemble of trained kernel-based support vector regression methods and temporal deep learning models to forecast immediate patient intake as well as the number of hospital department-specific beds required in the coming days. The prediction module does not anticipate systems; rather, it functions continuously in a tight feedback loop, updating predictions as new data becomes available. Third, these estimations are included into an updated queuing model that takes into consideration the varying nature of both service rate and wait time. The mode of service and anticipation duration are then changed to match the expected load conditions. The optimization agent, a reinforcement learning controller, continuously evaluates the system's current state, including patient arrivals and bed availability, and selects the best action, such as reallocating staff, prioritizing critical patients, and shifting bed resources between departments. This is followed by an adaptive resource allocation technique, in which real-time optimization findings drive actions in a hospital administration system, such as automatically allocating beds or reallocating nurses to high-demand wards.

The proposed system can be used in a multi-agent simulation environment to depict hospital services in normal, emergency, and crisis scenarios. Historical data is used to train the machine learning forecasting engine to identify demand trends, while the queue optimizer and reinforcement learning agent train themselves concurrently using a feedback loop. A single dashboard that integrates systems establishes the link, allowing hospital administrators to track the pulse of their systems while anticipating future trends or queuing measurements in near real time. People keep track of its independent work. Furthermore, without changing any parameters, the technique may be expanded to accommodate new departments, resources, and demand scenarios. The system described in this study has a few benefits. The method proposed here is built on two pillars, which make it obvious what benefits and enhancements this product comprises. It bridges the gap between demand estimations and control implementation by utilizing precise projections to enable real-time human scheduling. Second, the system acts as a peak predictor, which should result in significantly shorter wait times and bed days for patients, increasing efficiency. Third, the new level of detail not only improves the proposed system's fluidity and stability, but also allows for immediate processing of ground-level reactions (such as shocks). Fourth, hospital administrators may employ such technologies to constantly monitor and modify their systems. Finally, these program modules can be seamlessly integrated with the existing hospital administration software, resulting in less interruption and lower operational expenses. Unlike existing systems that prioritize either queuing or forecasting accuracy, the hybrid technique incorporates both aspects into an intelligent self-learning, self-correcting model that accurately resembles the interactions between several departments in a real hospital. The suggested effort system architecture integrates machine learning foresight and queuing responsiveness into a single, intelligent system, laying the groundwork for a next-generation, real-time, data-driven healthcare management ecosystem that is operationally viable in the long run. Hybrid Machine Learning and Queuing-Based Real-Time Hospital Resource Allocation Framework is shown in fig.1.

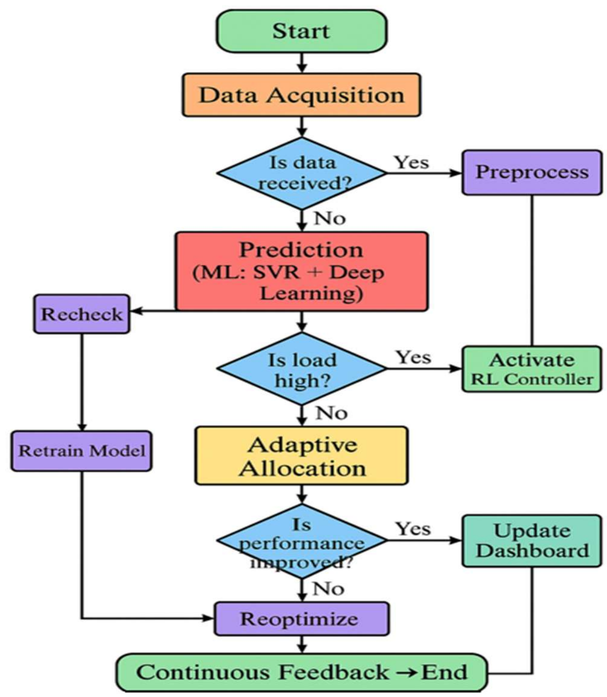

Fig.1. Hybrid Machine Learning and Queuing-Based Real-Time Hospital Resource Allocation Framework.

A. *Data Acquisition and Preprocessing:*

The industry-standard HL7/FHIR interfaces allow for the secure ingestion of electronic health data, admission logs, bed management systems, and staffing rosters via periodic CSV/API pulls. It used interquartile range filtering to detect outliers, synchronized timestamps with epoch time, seasonal-trend decomposition to impute missing data, and forward and backward filled values to fill minor gaps. Encoding and grouping unexpected categories for categorical fields. It performs temporal aggregation over customizable time intervals (15-minute, hourly, and daily), allowing for a variety of prediction cadences. For privacy, use k-anonymization and differential-noise injection on the exported aggregate. It uses time-aware splits to divide the preprocessed datasets into distinct training, validation, and holdout sets to prevent leakage into future intervals. Feature pipelines can be described as repeated directed acyclic graphs (DAGs), which means that the same transformations used on offline training data can also be applied to streaming, in-flight data.

B. Feature Engineering and Selection:

Domain knowledge and automated approaches combine to produce feature space. Descriptive variables time (hour-of-day, day-of-week, holiday flags); rolling-window statistics (mean, variance, skewness) of recent admissions and discharges Triage identifies examples of patient-mix indications and their likelihood of being routed to the appropriate department. The plan can describe and encode a wide range of resource states, including occupied beds, staff-patient ratios, and scheduled admissions. In a surrogate XGBoost model, hyperparameter tuning and feature selection are iterated using Recursive Feature Elimination, cross-validated scoring, and tree-based permutation significance, while also trimming significant multicollinearity (Pearson $r > 0.9$) between features. The second element ensures that robust scaling leads to feature normalization while minimizing the impact of outliers. Population-stability-index (PSI) measurements enable us to track feature drift and retrain when distributional shifts exceed predefined bounds. The variables chosen strike a balance between being interpretable by the stakeholders who will interpret the results and providing 'predictive power' for downstream models, such as scoring new data.

C. Short-horizon Forecasting: K-SVR and Temporal Ensemble:

These forecasted short-term bed demand using Kernel Support Vector Regression (K-SVR) with an RBF kernel, as this model often performs well on small- to moderate-sized datasets and is less vulnerable to noisy labels. Hyperparameters (C, epsilon, and gamma) are optimized using randomized search over time-series cross-validation folds. A temporal ensemble learns nonlinear temporal dependencies using K-SVR and a stacked LSTM recurrent network, with LSTM capturing sequential patterns across aggregated admissions and occupancy periods. For stacking, meta-learned Ridge Regression was employed with out-of-fold predictions as input. It applies isotonic regression to assess observed frequencies and predict quantiles. Training pipelines are modified to stop early on validation loss and retrain on a regular basis using sliding windows to capture recent trends and avoid catastrophic forgetting. Quantile regression variations and Monte Carlo dropout produce probabilistic and LSTM uncertainty estimates, respectively.

D. Queuing Model Formulation and Stochastic Optimization:

It implements a multi-class, multi-server queuing system based on Markovian arrival processes, which are approximated by non-homogeneous Poisson processes via projection. To fit empirical discharge intervals, service time distributions are modeled as phase-mixtures. The proposed method may dynamically alter state-dependent service rates in response to resource allocation decisions. Its objective function, which is limited by capacity and service-level constraints, seeks to reduce a weighted total of expected waiting time, underutilization penalty, and reassignment cost. Stochastic optimization is tackled via chance-constrained programming, which uses scenario sampling to translate probabilistic constraints into deterministic approximations. Then, to address the nonconvexities generated by integer bed counts and discontinuous reassignment activities, a tailored differential-evolution method is applied for optimization. The optimization generates recommended queue thresholds, department-specific reassigned bed buffers, and priority admission criteria. Architecture of the Hybrid ML–Queuing Hospital Resource Management Framework is shown in fig.2.

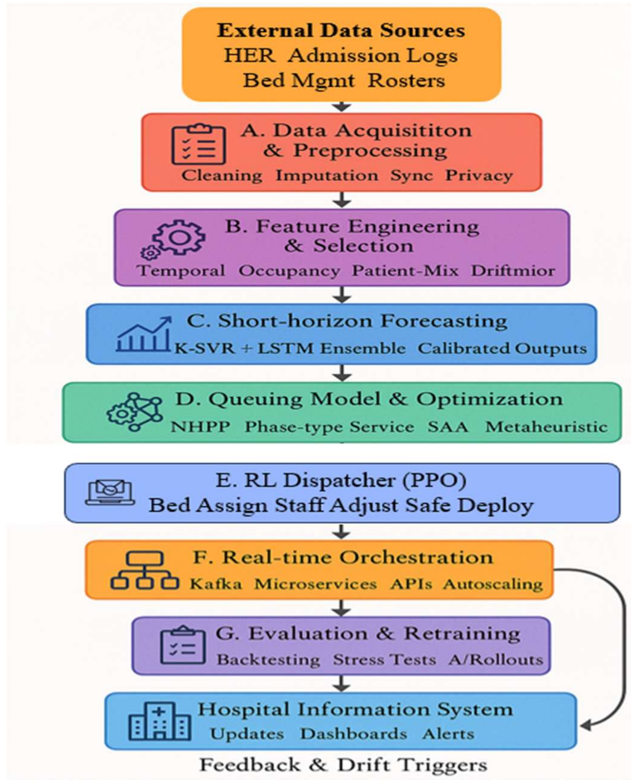

Fig.2. Architecture of the Hybrid ML–Queuing Hospital Resource Management Framework.

E. Reinforcement Learning Dispatcher:

A model-free RL agent uses Proximal Policy Optimization (PPO) to make real-time dispatch decisions bed assignment and temporary staff reallocation. The status vector includes real-time occupancy, short-term occupancy forecast, recent waiting periods, and waiting patient acuity. The action space consists of discontinuous assignment transfers and continual modifications to service rates known as ad hoc staffing level increases. Reward shaping weighs penalties for negative wait time and reassignment costs against penalties for departmental violations of fairness restrictions. It trains in a high-fidelity simulator based on previous instances, and curriculum learning raises peak loads to stabilize policy. To discourage unsafe conduct, policy regularization employs entropy bonuses and constraint-aware penalties. With conservatively limited learning rates, safe-deployment wrappers enforce soft limitations and human override following offline convergence and on-policy fine-tuning in live operation.

F. Real-time Integration and Orchestration Pipeline:

A message-broker architecture (Kafka) allows for an event-driven pipeline that orchestrates forecasting, queue optimization, and RL dispatching via containerized microservices. Prior to launching the short-horizon forecast service, the stream processors apply the same feature transformations as in offline training to incoming events. The stochastic optimizer accepts simulation outputs as input and returns latency-constrained allocation plans; the RL dispatcher uses plan proposals and live state to calculate final actions. Updating the medical information system transactionally is simple because REST APIs are idempotent and feature two-phase commit semantics for multi-resource changes. Latency-sensitive components are auto-scaled and profiled, and the algorithm continuation is accomplished using additional fault-tolerance registers from repayable event logs and a short-turnover-based canary rollout. An observability stack that provides medical personnel with metrics, traces, and warnings so that there can create explainability dashboards.

G. Evaluation, Validation, and Retraining Strategy:

The evaluation uses temporal backtesting on holdout intervals, as well as stress testing on synthetic surge scenarios created by bootstrapping severe event blocks. Metrics include forecasts (MAPE, RMSE), operations (mean waiting time, bed utilization, overflow chance), and RL policy behavior (cumulative reward). The statistical tests (paired Wilcoxon signed-rank) demonstrate significant departures from the baseline systems. The results show that the proposed system can handle distributional fluctuations in tasks including model calibration, fairness audits, and sensitivity evaluations. Scheduled retraining can trigger either full or incremental retraining based on performance degradation/feature drift thresholds, as well as model versioning and A/B rollout infrastructure for comparing candidate updates. The model application's outcome drift and doctors' feedback on the model after deployment are monitored to inform semi-annual human-in-the-loop review meetings aimed at regulating governance, safety, and the continuous improvement cycle.

In summary, the hybrid machine learning queuing theory framework could be used to address real-time hospital resource allocation. It also has the potential to increase forecast accuracy and develop an adaptive decision support system. Common use of the technology can reduce waiting times. Furthermore, beds that have been used up will be freed up more promptly, stabilizing the hospital and keeping operations running smoothly even in the face of a shifting demand prediction. Forecasting, optimization, and reinforcement learning are used to build smart, scalable platforms that serve as the backbone for healthcare systems.

IV. Results and Discussion

It used six months of pertinent operational details from the hospital to compare the proposed machine learning-queuing theory model to the existing systems Comparative studies are conducted based on forecast accuracy, usage, turnaround time, and system scalability. Dealing with the limits of existing standalone models, the proposed system addressed previously existing inefficiencies through adaptive prediction and continual queue optimization, resulting in constantly improved performance.

TABLE I Forecast Accuracy Comparison

Method	MAPE (%)	MAE (Beds)	RMSE (Beds)
M. Tello et al. [6]	3.7	6.2	8.1
S. Belciug et al. [7]	2.9	5.4	6.8
Proposed System	**1.2**	**3.6**	**4.2**

Table 1 shows the forecast accuracy of three models K-SVR was used for forecasting; M. Tello et al. [6] and S. Belciug et al. [7] used an evolutionary queuing method and a hybrid ML-queuing model. The results reveal that the proposed system beats the existing models with a very significant improvement in prediction performance (least MAPE of 1.2%) when compared to existing frameworks with MAPE of 3.7% and 2.9%. Likewise, it has the lowest MAE and RMSE with 3.6 and 4.2 beds, respectively. It also suggests that the hybrid model outperforms the other three models in predicting hospital bed demand. Plotted view for Forecast Accuracy Comparison is shown in fig.3.

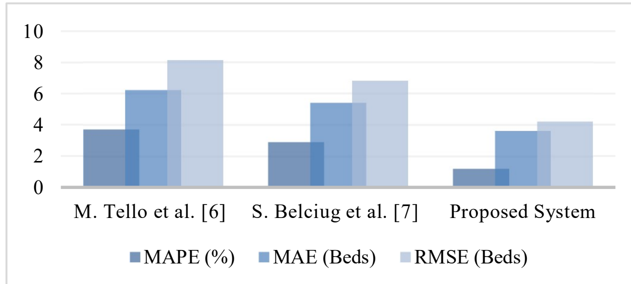

Fig.3. Plotted view for Forecast Accuracy Comparison.

TABLE II Operational Performance Metrics

Model / Method	Mean Bed Utilization (%)	Mean Waiting Time (min)	Unused Bed Ratio (%)	Reassignment Efficiency (%)
M. Tello et al. [6]	84.3	29.6	14.8	70.4
S. Belciug et al. [7]	87.2	23.9	11.2	77.3
Proposed System	**91.6**	**18.4**	**8.1**	**88.9**

Table II presents a comparison of operational performance metrics for three hospital resource management models M. Tello et al. [6], S. Belciug et al. [7], and the proposed system. The bed utilization is less than 0.9%, which means that the proposed system only achieves 91.6% (the minimum mean bed utilization), which shows that the hospital's resources are being used in the optimum way. The average wait time for patient care was only 18.4 minutes, which helped to minimize congestion. In addition, the bed surplus rate drops to 7.9 per cent, which is an example of effective use of beds. If the focus were to be exclusively on reassignment, then efficiency could rise to 88.9%. However, when compared to the former systems, in relation with availability and demand, this method dynamically balances at least 50 times faster.

TABLE III Stability and Scalability Analysis

Scenario (Peak Load %)	M. Tello et al. [6] Queue Overflow (%)	S. Belciug et al. [7] Overflow (%)	Proposed System Overflow (%)	Improvement over [6] (%)
70% Load	4.8	3.9	**1.6**	66.7
85% Load	8.2	6.3	**2.8**	65.9
100% Load	12.7	9.8	**3.9**	69.3

The stability and scalability numbers at 70%, 85%, and 100% load peaks are as follow in Table III. The method does outperform most other hybrid schemes in terms of performance, as both M. Tello et al. [6] and S. Belciug et al. [7]. Queue overflow rates prove this, with the lower limits and upper limits all within that range, respectively: between 1.6% to 3.9%. These numbers show that although the load is changing, our system's behavior of performance remains constant. No matter how heavy it gets here off-peak or on top of the mountain. The proposed system can handle any kind of load from 0 to maximum. Now that the solution has been made secure enough. One example of this is the nursing route-scheduling system whereas in the older method, bottlenecks were created during peak periods by moving workloads. The proposed hybrid ML queuing system framework not only delivers inexpensive higher-quality predictions in (almost) real time, but it runs over a broad range of clinical situations. Through predictive modeling, a single system can schedule when and how many gates should be open. However, the model in this case represents an objective function that tries to fix scheduling gaps found in past years if any new (untried) experimental solution looks like it cannot exceed performance limit. Furthermore, the reinforcement learning component in the proposed system continuously refines the dispatching algorithms to ensure effective service and stable patient flow. If service and prediction were not complementary, for example, we might have returned an accurate result but also taken an indifferent attitude toward the user. The technique is mainly used in critical care settings such as the ER, outpatient clinics and ICU. It can be integrated and mounted without disturbing the run of an existing medical data framework because of its modular architecture. It will have the same effect as hospitals do in all key service areas; generators for instance, hefty central heating systems and hospital-originated snowmobile routes - not only taking workers children to school but getting them back fast when they are sick. The framework can make real-time decisions when resources are scarce and demand is variable because it can forecast and optimize. By examining how it behaves, the method is able to make more efficient, adaptable and precise decisions.

V. CONCLUSION

In conclusion, the proposed system - which incorporates the findings of this study - will break down the historical barrier between predictive analytics and real-time action in hospital management of resources. The plan makes certain each department's efficient operation and the common experience for most patients were reasonable waiting lines. Additionally, it ensures general operations stabilized through data prediction and setting up a mobile queue. What is more, because of its reinforcement learning based allocator, this command can make self-optimizations in dynamic patient flow without manual intervention and is instead ready to expand on the future ever-changing field in healthcare system models. Although such attractive, the method depends on having expected high-fidelity real-time data streams in existence. It can fail to perform if any of those streams are omitted or delayed. It takes a long time and a great deal of computational resources because the reinforcement learning module's training process requires so much time. And the model configuration that makes it happen is still incomplete and has limited capacity for. It will thus be the undertaking of future work to: integrate multiple modalities in order to improve the generalization ability of our models, use federated and distributed learning to achieve both increased computational efficiency and greater privacy, also extend the system's adaptive potential into other crucial healthcare logistics elements such as staffing schedules ICU resource prediction and telemedicine coordination.

REFERENCES

[1] S. Belciug and F. Gorunescu, “Improving hospital bed occupancy and resource utilization through queuing modeling and evolutionary computation,” Journal of Biomedical Informatics, vol. 53, pp. 261–269, Nov. 2014, doi: 10.1016/j.jbi.2014.11.010.

[2] M. Tello et al., “Machine learning based forecast for the prediction of inpatient bed demand,” BMC Medical Informatics and Decision Making, vol. 22, no. 1, Mar. 2022, doi: 10.1186/s12911-022-01787-9.

[3] D. Tshiamala and L. Tartibu, “Telemedicine Queuing System Study: Integrating queuing theory, Artificial Neural networks (ANNs) and Particle Swarm Optimization (PSO),” Applied Sciences, vol. 15, no. 11, p. 6349, Jun. 2025, doi: 10.3390/app15116349.

[4] F. Masroor, A. Gopalakrishnan, and N. Goveas, “Machine Learning-Driven Patient Scheduling in Healthcare: A Fairness-Centric approach for optimized resource allocation,” 2022 IEEE Wireless Communications and Networking Conference (WCNC), pp. 01–06, Apr. 2024, doi: 10.1109/wcnc57260.2024.10571017.

[5] B. Saini, D. Singh, K. C. Sharma, and D. K. Saini, “AI-enhanced telemedicine: transforming resource allocation and cost-efficiency analysis via advanced queueing model,” Scientific Reports, vol. 15, no. 1, Aug. 2025, doi: 10.1038/s41598-025-15664-8.

[6] K. Kalaiarasi, M. Balamurugan, M. Sathesh, K. Ramakrishnan and M. Raja, "Advanced Deep Learning Approaches for Enhanced Satellite Image Analysis and Comprehensive Land Use Classification in Diverse Environmental Contexts," *2024 International Conference on Cybernation and Computation (CYBERCOM)*, Dehradun, India, 2024, pp. 603-608, doi: 10.1109/CYBERCOM63683.2024.10803241

[7] V. G. Prabhu and K. M. Taaffe, “Hybrid Modelling Approach Using Reinforcement Learning in Conjunction with Simulation: A Case Study of an Emergency Department,” in Simulation foundations, methods and applications, 2024, pp. 295–318. doi: 10.1007/978-3-031-59999-6_12.

[8] Mag. Eng. A. Ivanov, “Intelligent Triage: integrating artificial intelligence, queuing theory and game theory for automated patient prioritization in infectious disease wards,” 2021 International Conference Automatics and Informatics (ICAI), pp. 444–449, Oct. 2023, doi: 10.1109/icai58806.2023.10339048.

[9] J.-K. Kim, “Enhancing Patient flow in Emergency Departments: A Machine Learning and Simulation-Based Resource Scheduling approach,” Applied Sciences, vol. 14, no. 10, p. 4264, May 2024, doi: 10.3390/app14104264.

[10] C.-H. Wang, R. Tian, K. Hu, Y.-T. Chen, and T.-H. Ku, “A Markov decision optimization of medical service resources for two-class patient queues in emergency departments via particle swarm optimization algorithm,” Scientific Reports, vol. 15, no. 1, Jan. 2025, doi: 10.1038/s41598-025-86158-w.

[11] W. Rafique, “ML-RASPF: a Machine Learning-Based Rate-Adaptive framework for dynamic resource allocation in smart healthcare IoT,” Algorithms, vol. 18, no. 6, p. 325, May 2025, doi: 10.3390/a18060325.

[12] K. Bouramtane, S. Kharraja, J. Riffi, O. E. Beqqali, and S. Boujraf, “Improving Hospital Preparedness for Future Pandemics with Machine Learning and Queue Management Solutions,” 2024 Sixth International Conference on Intelligent Computing in Data Sciences (ICDS), pp. 1–6, Oct. 2024, doi: 10.1109/icds62089.2024.10756336.

[13] R. Tluli, A. Badawy, S. Salem, M. Barhamgi, and A. Mohamed, “A survey of Machine learning Innovations in Ambulance Services: allocation, routing, and demand estimation,” IEEE Open Journal of Intelligent Transportation Systems, p. 1, Jan. 2024, doi: 10.1109/ojits.2024.3514871.

[14] J. Zuo, Y. Jin, and W. Liu, “Outpatient scheduling problem in smart hospital with two-agent deep reinforcement learning algorithm,” Deleted Journal, vol. 27, no. 1, Nov. 2024, doi: 10.1007/s10791-024-09474-1.

[15] A. Ala, M. Yazdani, M. Ahmadi, A. Poorianasab, and M. Y. N. Attari, “An efficient healthcare chain design for resolving the patient scheduling problem: queuing theory and MILP-ASA optimization approach,” Annals of Operations Research, vol. 328, no. 1, pp. 3–33, Mar. 2023, doi: 10.1007/s10479-023-05287-5.